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Remotely activated portable hand tool

Abstract

The present disclosure provides hand-held, battery-powered tools for performing operations such as crimping and cutting, and having a remote monitoring and control systems. A frame of the tool supports a working head, a battery, a motor connected to the battery, a controller and a camera.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present disclosure is a continuation of U.S. application Ser. No. 15/249,371 filed Aug. 27, 2016, now U.S. Pat. No. 11,440,153, which claims benefit from U.S. Provisional Application No. 62/210,572 filed Aug. 27, 2015, the entire contents of both are incorporated herein by reference.

BACKGROUND

Field

(1) The present disclosure relates generally to portable power tools, and more specifically to portable power tools with systems for remote monitoring and operation of the working head of the tool.

Description of the Related Art

(2) Most portable power tools are hand held tools that use electric motors to drive a working head to perform various tasks, such as crimping, drilling, shaping, fastening, grinding, polishing, heating, etc. There is a segment of the portable tool product market that incorporate a hydraulic pump to enable the working head to apply a relatively large amount of force or pressure for a particular task. Such tools may operate with a hydraulic pump actuated by a battery powered electric motor. Battery powered hydraulic power tools are employed in numerous applications to provide an operator with a desired flexibility and mechanical force. For example, tools may need substantial force to crimp large power connectors onto large conductors, e.g., #8 conductors and larger. As another example, tools may need substantial force to cut large conductors, e.g., #8 conductors and larger.

(3) There are certain environments where activation of portable, battery powered tools, including portable, battery powered hydraulic power tools, is best performed remotely to provide additional operator safety. For example, when working with high current electric lines, activating a portable, battery powered tool remotely permits an operator to avoid exposure to such high currents.

SUMMARY

(4) The present disclosure provides hand-held, battery-powered tools having a remote monitoring and remote control systems. A frame of the tool supports a working head, a battery, a motor

connected to the battery, a controller and in some embodiments a camera. In an exemplary embodiment, the portable tool includes a frame configured to be hand held, a working head secured to the frame and configured to perform at least one action, a camera secured to the frame, and a controller for controlling the operation of the working head and the camera such that a video of a working area can be presented to an operator for viewing the operation of the working head. In another exemplary embodiment, a portable tool system is provided. In this embodiment, the portable tool system includes a portable tool, similar to the tool above, and a computing device, operatively connected to the tool, wherein a video of a working area can be presented to an operator on the computing device for viewing the operation of the working head.

(5) In another exemplary embodiment, the portable tool system includes a frame configured to be hand held, a working head secured to the frame and configured to perform at least one action, a controller for controlling the operation of the working head, and a wireless module coupled to the controller and configured to exchange information between the controller and a wireless computing device, and to permit the wireless computing device to remotely control the operation of the working head. A portable tool system using the tool in this embodiment is also provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate presently preferred embodiments of the invention, and together with the general description given above and the detailed description given below, serve to explain the principles of the invention. As shown throughout the drawings, like reference numerals designate like or corresponding parts.

(2) FIG. 1 is a perspective view of a battery operated hydraulic electrical connector crimping tool according to the present disclosure.

(3) FIG. 2 is a block diagram of components of the tool shown in FIG. 1.

(4) FIG. 3 is a block diagram of an exemplary embodiment of a power supply module included in the tool of FIG. 1;

(5) FIG. 4 is a circuit diagram of an exemplary embodiment of a controller according to the present disclosure;

(6) FIG. 5 is a circuit diagram of an exemplary embodiment of a flash memory module according to the present disclosure;

(7) FIG. 5A is a circuit diagram of another exemplary embodiment of a flash memory module according to the present disclosure;

(8) FIG. 6 is a circuit diagram of an exemplary embodiment of a real time clock and calendar (RTCC) module according to the present disclosure;

(9) FIG. 7 is a circuit diagram of an exemplary embodiment of a motor current sensor according to the present disclosure;

(10) FIG. 8 is a circuit diagram of an exemplary embodiment of a pressure sensor according to the present disclosure;

(11) FIG. 9 is a circuit diagram of an exemplary embodiment of a battery voltage monitor according to the present disclosure;

(12) FIG. 10 is a circuit diagram of an exemplary embodiment of a wireless communication module circuit according to the present disclosure;

(13) FIGS. 11A and 11B are circuit diagrams that together form of an exemplary embodiment of a USB interface module according to the present disclosure;

(14) FIG. 12 is a circuit diagram of an exemplary embodiment of a USB power supply circuit according to the present disclosure;

(15) FIG. 13 is a circuit diagram of an exemplary embodiment of a motor drive module according

to the present disclosure;

(16) FIG. **14** is a circuit diagram of an exemplary embodiment of a LED module according to the present disclosure;

(17) FIG. **15** is a circuit diagram of an exemplary embodiment of a buzzer module according to the present disclosure;

(18) FIGS. **16-18** are screen shots on an exemplary embodiment of a hand held computing device for monitoring the operation of the tool of FIG. **1**;

(19) FIG. **19** is an exemplary embodiment of a hand held computing device displaying a video of a working area relative to a working head of the tool of FIG. **1**; and

(20) FIG. **20** is a block diagram of an exemplary embodiment of a system for monitoring the operation of the tool of FIG. **1**.

DETAILED DESCRIPTION

(21) The present disclosure provides portable hand tools including systems for remote monitoring and operation of a working head of the hand tool. Referring to FIG. **1**, an exemplary embodiment of a portable hand tool **10** is shown according to the present disclosure. Although the present disclosure describes the hand tool as a portable, hand held, battery operated, hydraulic crimping tool, it should be understood that the tool of the present disclosure is not limited to such crimping tools. Features of the portable hand tool of the present disclosure could also be used in other types of tools, such as a battery operated, hydraulic cutting tools or any other suitable type of battery operated tool. In addition, any suitable size, shape or type of elements or materials can be used to form the shape of the tool frame. For ease of description, the portable, hand held, battery operated, hydraulic crimping tool shown and described herein will be referred to as the “tool.”

(22) Referring to FIGS. **1** and **2**, the tool **10** according to an exemplary embodiment of the present disclosure generally includes a frame **12** and a working head **14**. The frame **12** includes a pump **16**, a motor **18**, a battery **20**, a fluid reservoir **22**, a controller **24** and a hydraulic drive conduit system **28**. The frame **12** includes a main body **13** and a handle **40** that form a pistol-like shape. However, the frame **12** could be in any suitable shape, such as an in-line shape.

(23) The pump **16**, motor **18**, fluid reservoir **22**, controller **24**, and hydraulic drive conduit system **28** are located within the main body **13** of the frame **12**. The tool **10** may also include a camera **26** mounted to the frame **12** and oriented to provide a video of a working area of the working head **14**, as will be described in more detail below. The working head **14** includes a frame section **30** and a ram **32**. The frame section **30** may be connected to the front end of the frame **12** and fixed or locked in position, or the frame section **30** may be rotatably connected to the frame **12**. The ram **32** is movably connected to the frame section **30** of the working head **14**. In the exemplary embodiment shown, the frame section **30** and the ram **32** are adapted to removably receive conductor crimping dies (not shown) at a conductor receiving area **34**.

(24) The ram **32** is adapted to move forward and backward as indicated by arrow **36**. The hydraulic drive conduit system **28** is connected between the pump **16** and the rear end of the ram **32**.

Hydraulic fluid pumped by the pump **16** through the hydraulic drive conduit system **28** and against the rear end of the ram **32** causes the ram **32** to move forward toward a distal end of the working head **14**. The tool **10** preferably includes a spring (not shown) which is adapted, as is known in the art, to return the ram **32** to its rearward (or home) position when hydraulic fluid pressure is removed from the rear end of the ram **32**. In the exemplary embodiment shown, the ram **32** has a rear end diameter of about 2 inches. However, the diameter of the rear end of the ram could have any suitable size or shape for functioning as a hydraulic fluid contact surface. In the exemplary embodiment shown, the ram **32** is adapted to move a distance “D” between its home position and its forward position which is towards the distal end of the working head **14**. The distance “D” can be any distance suitable to perform the desired action of the working head **14**, here the desired crimping action. For example, the distance “D” could be between about 1 inch and about 2 inches, and in a preferred embodiment about 1.7 inches.

(25) The handle **40** of the frame **12** may include a hand guard **42** to protect an operators hand while operating the tool **10**. The handle **40** includes one or more operator controls, such as trigger switches **44** and **46**, that can be activated by an operator by, for example, pressing the trigger switches. The operator controls, e.g., trigger switches **44** and **46**, are operably coupled to the controller **24**, as will be described below. As shown in FIG. 1, the battery **20** is removably connected to the bottom of the handle **40**. In another embodiment, the battery **20** could be removably mounted or connected to any suitable position on the frame **12**. In another embodiment, the battery **20** may be affixed to the tool **10** so that it is not removable. The battery **20** is preferably a rechargeable battery, such as a lithium ion battery, that can output a voltage of at least 16 volts, and preferably in the range of between about 16 VDC and about 24 VDC. In the exemplary embodiment shown in FIG. 1, the battery **20** can output a voltage of about 18 VDC.

(26) The motor **18** is coupled to the battery **20** and the controller **24**, and its operation is controlled by the controller **24**, which will be described in more detail below. Generally, the motor **18** is adapted to operate at a nominal voltage corresponding to the voltage of the battery **20**, i.e., between about 16 volts and about 24 volts. For example, if the battery **20** is adapted to output a voltage of about 18 volts, then the motor **18** would be adapted to operate at a voltage of about 18 volts. Under a no-load condition, such a motor **18** can operate at about 19,500 rpm with a current of about 2.7 amps. At maximum efficiency, the motor **18** can operate at about 17,040 rpm with a current of about 18.7 amps, a torque of about 153 mN-m (1560 g-cm), and an output of about 273 W. An example of such an 18-volt motor **18** may be a RS-775WC-8514 motor, manufactured by Mabuchi Motor Co., Ltd. of Chiba-ken, Japan. However, as noted above, any suitable type of motor adapted to operate above a 16 V nominal voltage could be used. For example, the motor may be a RS-775VC-8015 motor, also manufactured by Mabuchi Motor Co., Ltd., which has a nominal operating voltage of about 16.8 volts. As another example, the motor may be a motor adapted to operate at a 24 V nominal voltage. The output shaft of the motor **18** is connected to the pump **16** by a gear reduction or gearbox **48**. Any suitable type of gear reduction assembly could be provided.

(27) Referring again to FIG. 2, the tool **10** may include a poppet valve **50** connected to the hydraulic drive conduit system **28**. The poppet valve **50** is adapted to open when the conduit system **28** reaches a predetermined pressure, such as between about 8000 and about 11,000 psi. When the poppet valve opens, hydraulic fluid being pumped by the pump **16** can exit the conduit system **28** and return to the fluid reservoir **22**. The poppet valve **50** can be adapted to generate an audible sound when it opens. This audible sound can signal to the operator that the tool **10** has reached its maximum predetermined hydraulic pressure and, thus, the action of the working head **14**, e.g., crimping action, is ready to be triggered.

(28) In the exemplary embodiment shown in FIG. 2, the controller **24** is adapted to sense a current drop of electricity to the motor **18**. When the poppet valve **50** opens, resistance to rotation of the motor **18** is reduced such that the motor draws less current. The controller **24** senses this current drop via the current sensor **120** (seen in FIG. 7), and automatically deactivates the motor **18** for a predetermined period of time. In a preferred embodiment, the predetermined period of time is between about 2 seconds and about 3 seconds. However, any suitable predetermined period of time could be set. In an alternate embodiment, the controller **24** could be adapted to deactivate the motor **18** until a reset button or reset-like procedure is performed by the operator. With this type of system, an operator can sense via a tactile sense, that the motor **18** and pump **16** have stopped and would not need to rely on an audible signal being heard or a visual signal from an LED indicator light positioned on the tool **10**. More detailed information of this exemplary embodiment of the tool **10** and its operation can be found in U.S. Pat. No. 7,165,439 entitled "Battery Powered Hydraulic Tool" which is incorporated herein by reference.

(29) Referring to FIG. 3, a circuit diagram of a power supply module **150** is shown. As noted, the tool **10** can work with a battery of at least 16 VDC, which energizes both the power supply module **150** and the motor **18**. The power supply module **150** supplies power for the components of the

controller **24**, such as the microcontroller, the flash memory, the RTCC, and the Sensors and the Modules, described below, by reducing the battery voltage to a voltage suitable to power the components of the controller **24**. As an example, the power supply module **150** regulates the battery voltage, e.g., 18 VDC, to about 3.3V using a low dropout (LDO) voltage regulator U**8**. An example of a suitable voltage regulator is the TPS70933 LDO voltage regulator, manufactured by Texas Instruments. The LDO voltage regulator U**8** is enabled by an operator activating an operator control, e.g., pressing trigger switch **44** or **46**, causing the microcontroller **100** to take control of the power supplied to the tool **10**.

(30) The enable pin of the LDO voltage regulator U**8** enables the voltage regulator. The enable pin is normally pulled to ground using a 10K resistor R**3** which disables the LDO voltage regulator U**8** causing the microcontroller **100** to go into a power down state, as described below. When a trigger switch **44** or **46** is closed, the enable pin of the LDO voltage regulator U**8** is driven to a high voltage through the R**1**-R**3** voltage divider network. A high voltage on the voltage regulator U**8** enable pin (EN) enables the LDO voltage regulator U**8**, which then powers the microcontroller **100** via the VCC power and turns the microcontroller ON. Once the microcontroller **100** is ON, the microcontroller takes control of the power supply module by enabling the PWR_CTRL_MAIN line, thus maintaining a high voltage on the enable pin of the LDO voltage regulator U**8** even when the trigger switch **44** or **46** is not closed. If the tool **10** is kept idle for more than a particular predefined time, preferably a time in the range of 3-180 minutes, the microcontroller **100** disables the PWR_CTRL_MAIN line, thus removing the high voltage on the enable pin of the voltage regulator U**8** so that the LDO voltage regulator U**8** turns OFF until the trigger switch **44** or **46** is closed again, and power to the components of the controller **24** is removed.

(31) Continuing to refer to FIG. **3**, when the LDO voltage regulator U**8** is enabled and the components of the controller **24** are powered using VCC power, a further or continued pressing of the trigger switch **44** or **46** will then cause the microcontroller **100** to actuate the motor **18** thus turning the motor **18** ON. More specifically, the trigger switch **44** or **46** provides battery **20** voltage to the R**4**-R**13** resistor network which causes a high voltage at the base of n-channel MOSFET M**10**, which turns the MOSFET M**10** ON, and causes the OPERATOR_CTRL to be low which triggers the microcontroller **100** to turn ON the motor **18**. An example of a suitable MOSFET is the DMN601WK series MOSFET manufactured by Diodes, Inc. Once MOSFET M**10** turns ON, the OPERATOR_CTRL line is pulled to ground (Low Signal), through the MOSFET M**10**. On release of trigger switch **44** or **46**, the MOSFET turns OFF and the OPERATOR_CTRL line is pulled back to high through resistor R**11**. The OPERATOR_CTRL line is connected to microcontroller interrupt (seen in FIG. **4**) so that the low signal causes the microcontroller **100** to turn the motor **18** on.

(32) Referring to FIG. **4**, an exemplary block diagram of the controller **24** of the present disclosure is shown. In this exemplary embodiment, the controller **24** may include a microcontroller **100**, a wireless communication module **104**, a USB module **106**, a flash memory module **108**, and a real time clock and calendar (RTCC) **110**. The controller **24** may also include one or more of the following sensors: current sensor **120**, pressure sensor **122**, battery monitor sensor **124**, hydraulic temperature sensor **126**, and printed circuit board (PCB) temperature sensor **128**. Collectively, the sensors above are also referred to herein as “Sensors.” The controller **24** may also include one or more of the following modules: power control module **130**, motor drive module **132**, LED module **134**, buzzer module **136**, sensor power control module **138**, and a camera module **140** configured to control the operation of the camera **26**. Collectively, the modules above are also referred to herein as “Modules.” Power to the components of the controller **24** is provided by the power supply module **150**.

(33) The microcontroller **100** controls the operation of the tool **10**. The microcontroller **100** includes a 4 KB EEPROM, three timers, a 16 kilobyte static RAM, a general purpose input/output (GPIO) interface **100a**, an analog to digital converter (ADC) interface **100b**, hardware interrupt interface **100c**, universal asynchronous receivers/transmitters (UARTs) **100d**, a serial peripheral

interface (SPI) **100e**, an I2C interface **100f** and a serial camera interface **100g**. The microcontroller **100** is preferably an 8-bit microcontroller, such as the ATmega 1284 8-bit microcontroller manufactured by Atmel Corp. The microcontroller **100** interfaces with a 11.0592 MHz crystal, which permits faster processing and substantially error free serial communications.

(34) The general purpose input/output (GPIO) port **100a** of the microcontroller **100** controls one or more of the Modules included in the controller **24**. The microcontroller **100** disables one or more Sensors via the Sensor Power Control Module **138** when those Sensors are not in use, thereby conserving battery power. A pico-power mode of the microcontroller **100** helps the tool **10** work more efficiently as the microcontroller will use less power in low power modes, described below. The microcontroller **100** includes a number of separate external interrupt options, which allows the microcontroller to run in sleep mode most of the time.

(35) The wireless communication module **104** and the USB module **106** provide external connectivity to the tool **10** via the UART interfaces **100d** of the microcontroller **100**. An external computing device, such as a personal computer, smartphone, mobile phone, tablet, or PDA, can communicate with the tool **10** via the wireless communication module **104** and/or the USB module **106**. The wireless communication module **104** and the USB module **106** may use standard communication protocols for communications with the external computing device. When the wireless communication module **104** connects to an external computing device, the wireless communications module **104** may generate an interrupt to the hardware interrupt interface **100c** of the microcontroller **100** to cause the microcontroller to awake from sleep mode and communicate with the external computing device.

(36) The controller **24** also includes a flash memory **108** to store firmware code and data related to the operation of the tool **10**. Preferably, the flash memory **108** is an 8 MB flash memory, and is interfaced with the microcontroller **100** via the Serial Peripheral Interface (SPI) **100e**. The controller **24** includes a real time clock and calendar (RTCC) **110** used to generate the current date and time for timestamps of certain events such as operation of the tool **10**. The RTCC **110** is interfaced with the microcontroller **100** via the I2C interface **100f**. The RTCC **110** also provides interrupts via the hardware interrupt interface **100c** to the microcontroller **100** to initiate a countdown to a sleep mode and power-down mode. In some embodiments, the RTCC interrupt is generated once every minute, but other periods may be used.

(37) FIGS. 5 and 5A illustrate alternative embodiments of the flash memory **108**. The flash memory **108** is provided to store firmware and tool information. Tool information includes information about the tool's performance, actions by the working head, readings from the sensors, information entered by any computing devices, and any other information about the tool **10**. Examples of such tool information include date and time information for various events, the cumulative number of working head actions, battery status, and working head action history. In one embodiment, each successful crimp action would utilize about 256 bytes of flash memory space to store information about such crimping action.

(38) The flash memory chip U12 may be about 8 MB in size and is interfaced with the microcontroller **100** via an SPI interface **100e** embedded in the microcontroller **100** to ensure high speed data transfer between the flash memory and the microcontroller. The SPI interface **100e** includes pins SPI_MISO, SPI_MOSI, SPI_CLK, and SPI_SS1.

(39) In FIG. 5, the flash memory **108** can be disabled by the microcontroller **100** using an SPI slave select pin (SPI_SS1) which is connected to the Chip Select (CS) pin of flash memory **108**. In some embodiments, multiple slave select pins are used to select a particular one of multiple flash memory chips to be addressed through the SPI interface **100e**.

(40) In another embodiment shown in FIG. 5A, the flash memory **108** can be disabled by microcontroller **100** using a MEMORY_CTRL line from the controller **100**, and the MOSFET circuit comprising MOSFET M18 and resistors R23 and R50. When the MEMORY_CTRL line is asserted, the GND pin of the flash memory **108** is connected to ground via MOSFET M18. When

the MEMORY_CTRL line is not asserted, the GND pin of the flash memory **108** is isolated from ground by MOSFET **M18**. An example of a suitable MOSFET is the DMN601WK series MOSFET manufactured by Diodes, Inc.

(41) To conserve the use of battery power, the tool **10** can operate in one or more modes. For example, the tool **10** can be configured to operate in 3 modes; an active mode; a sleep mode and a power down mode.

(42) In the active mode, the tool **10** is able to perform its functions. For example, the tool **10** under the control of the microcontroller **100** can cause the working head **14** to perform its designed function, e.g., a crimping function, a cutting function, etc. The tool Sensors and Modules are also active. In addition, the tool **10** can communicate with external computing devices via the wireless communication interface **104** and the USB interface **106**. In the active mode, if the tool **10** is idle for an operator-defined period of time, e.g., 2 minutes, which is counted by the RTCC **110** as described below, the microcontroller **100** can cause the tool **10** to switch from the active mode to the sleep mode.

(43) In the sleep mode, the microcontroller **100** turns OFF all Modules and Sensors, except the wireless communication interface module **104**, and turns OFF other elements of the microcontroller **100**. During the sleep mode, the power supply module **150**, seen in FIG. 3, will supply some power to the tool **10**, e.g., to the microcontroller **100** while in its sleep mode and the wireless communication module **104** which will continue to advertise itself for connection to an external computing device. Since the wireless communication module **104** is active during sleep mode, the tool finder operation, described below, will be active and functional. If during sleep mode an external computing device with wireless communication functionality, e.g., Bluetooth® functionality, requests to connect to the tool **10**, the wireless communication module **104** will send an interrupt to the microcontroller **100** via the hardware interrupt interface **100c**. In response to the interrupt, the microcontroller **100** will switch the tool **10** to the active mode and respond to communications from the external computing device, e.g., a smartphone.

(44) If the tool **10** is idle, e.g., not being used, in sleep mode for an operator-defined time, the microcontroller **100** will turn the tool **10** OFF to the power down mode. In the power down mode, the tool **10** is turned off to a lower power state than sleep mode, such that the Modules and Sensors, the LDO voltage regulator **U8**, and the microcontroller **100** are OFF, and operations, such as the tool finder operation and remote operation of the tool **10** do not work without first turning on the tool **10**. When the tool is in power down mode, the operator can press the trigger switch **44** or **46** to switch the tool **10** to the active mode. Preferably, when the battery **20** is inserted to the tool **10**, the tool will be in the power down mode.

(45) Referring to FIG. 6, the real time clock and calendar (RTCC) Module **110** maintains accurate time and date information. An example of an RTCC chip is the MCP79410N, manufactured by Microchip Technology, Inc., Chandler, Arizona, United States. The RTCC Module **110** provides a time and date stamp for each stored working head action, e.g., each crimp action, to track the history of the tool and its use by operators. The RTCC Module **110** provides an interrupt at a predetermined period, e.g., once each minute, which is counted to determine the idle time of the tool **10** in units of the predetermined period, for sleep mode and power down mode operation. When the tool **100** is in active mode, and the RTCC **110** determines the tool has been idle for a first predetermined time, it communicates with the microcontroller **100** via I2C interface **100f**, and sends an interrupt to the microcontroller **100** via the hardware interrupt interface **100c** embedded in the microcontroller **100**. The microcontroller **100** is programmed to enter a sleep mode after the tool **100** is idle for a first predetermined period of time that is counted by the RTCC, as described above. When the microcontroller **100** is in sleep mode, an external interrupt may cause the microcontroller **100** to return to active mode and respond. As a result of being in sleep mode in the intervening period, the microcontroller **100** conserves battery power.

(46) When the tool **100** is in sleep mode, and the RTCC **110** determines the tool has been idle for a

second predetermined time, it communicates with the microcontroller **100** via I2C interface **100f**, and sends an interrupt to the microcontroller **100** via the hardware interrupt interface **100c** embedded in the microcontroller **100**. The microcontroller **100** is programmed to enter a power down mode after the tool **100** is idle for a second predetermined period of time that is counted by the RTCC, as described above. When the microcontroller **100** is in power down mode, the operator can press the trigger switch **44** or **46** to switch the tool **10** to the active mode. As a result of being in power down mode in the intervening period, the microcontroller **100** conserves battery power.

(47) Sensors are used to monitor and analyze different parameters of the tool **10**. The Sensors are interfaced with the microcontroller **100** via an analog to digital converter (ADC) **100b** (seen in FIG. **4**) embedded in the microcontroller **100**. The Sensors permit the microcontroller **100** to monitor and analyze the operations of the tool **10** and to make the tool **10** safe. In one embodiment, the sensors include a motor current sensor **120**, a hydraulic pressure sensor **122**, a battery voltage monitor **124**, a hydraulic temperature sensor **126** and a PCB temperature sensor **128**.

(48) Referring to FIG. **7**, the motor current sensor **120** is used to monitor the motor current. The motor current sensor **120** makes use of a current sensor chip U**13**, which may be the ACS-758 Hall effect current sensor manufactured by Allegro MicroSystems, LLC. The motor current sensor **120** provides an analog voltage corresponding to the current flowing through the motor **18** received through the IP+ and IP- pins of the current sensor U**13**. A low pass filter comprising resistor R**37** and capacitor C**51** smooths the voltage waveform on the VOUT line and to produce the filtered signal on the CURRENT_SENSOR line that is coupled to the microcontroller **100**. Based on the CURRENT_SENSOR signal, the microcontroller **100** is able to detect valid operation of the working head **14**, e.g., a valid crimp operation. The motor current sensor **120** can be disabled by microcontroller **100** to conserve power. The sensor power control module **138** drives the CURRENT_SENSOR_PWR_CTRL line which is coupled to the MOSFET M**12** and through the voltage divider comprising resistors R**21** and R**39**. When the CURRENT_SENSOR_PWR_CTRL line is driven high, the MOSFET **12** pulls the GND input of the current sensor chip U**13** to ground thereby allowing power to be delivered to the current sensor chip U**13**. When the CURRENT_SENSOR_PWR_CTRL line is driven low, the MOSFET **12** is turned off and current sensor chip U**13** is turned off. An example of a suitable MOSFET is the DMN601WK series MOSFET manufactured by Diodes, Inc.

(49) Referring to FIG. **8**, in instances where the tool **10** includes a hydraulic drive for added force, the hydraulic pressure sensor module **122** is used to monitor the pressure of the hydraulic fluid in the hydraulic drive conduit system **28**. The pressure sensor module **122** includes a pressure transducer connected through connector CON**4** to the printed circuit board (PCB). Line **1** of CON**4** is connected to VCC, Line **2** of CON**4** is the pressure sensor output, Line **3** of CON**4** is the GND pin, and Line **4** of CON**4** is the connected directly to GND.

(50) An example of a suitable pressure transducer is the pressure sensor welded on the modified MPM8-5 3/8-24 UNF port, manufactured by ICS-NH, Germany. The pressure transducer connected to the connector CON**4** converts the hydraulic fluid pressure to an analog voltage. The output of the transducer is linear with the hydraulic pressure generated. The pressure transducer is calibrated at 90% of VCC and corresponds to the maximum rated pressure of the tool. For example, the tool may be rated for 11,500 PSI. The pressure sensor module also includes a low pass filter comprising resistor R**29** and capacitor C**11**, which is provided to smooth the output of the pressure transducer. This filtered signal is then passed to the microcontroller's ADC **100b** via the PRESSURE_TRANSDUCER line. The pressure sensor module **122** can be disabled by microcontroller **100** to conserve power using the PRESSURE_TRANSDUCER_PWR line, which is connected to the sensor power control module **138**. The signal from the sensor power control module **138** is driven onto the PRESSURE_TRANSDUCER_PWR line to the MOSFET circuit comprising MOSFET M**14** and resistors R**27** and R**28**. When the PRESSURE_TRANSDUCER_PWR signal is high, Line **3** is pulled to ground, enabling the

pressure sensor. When the PRESSURE_TRANSDUCER_PWR signal is low, Line 3 is not pulled to ground, disabling the pressure sensor. An example of a suitable MOSFET is the DMN601WK series MOSFET manufactured by Diodes, Inc.

(51) Referring to FIG. 9, the battery monitor **124** scales the battery voltage through R8-R9 voltage divider network and zener diode D1, and is provided to the microcontroller's ADC **100b** via the BAT_MONITOR line. The microcontroller **100** under program control (e.g., firmware) takes the voltage reading from the BAT_MONITOR line and compares that voltage with a stored state of charge curve to calculate the voltage status of the battery **20**.

(52) If an 18V DC rated Li-Ion battery **20** is used in the tool **10** and is fully charged, the battery may provide a voltage of about 20V DC, which is higher than the rated voltage and decreases according to the discharge of the battery **20**. To monitor the battery voltage, a state of charge curve is first developed for each battery that is compatible with the tool **10**. To develop the state of charge curve, the battery **20** is loaded with a constant load, and state of charge curve is plotted over time. From this analysis, a relation between the battery charge and the battery voltage can be developed. These relationships are stored so that the voltage status can be determined by the battery monitor.

(53) The battery voltage monitor module **124** can be disabled by microcontroller **100** to conserve power using the BAT_MOTOR_PWR_CONTROL line, which is driven by the sensor power control module **138**. When the BAT_MOTOR_PWR_CONTROL signal is high, the MOSFET M16 is turned on through voltage divider R14-R16 and MOSFET M17 is turned off. When the BAT_MOTOR_PWR_CONTROL signal is low, the MOSFET M16 is turned off through voltage divider R14-R16 and MOSFET M17 is turned on by the battery voltage through resistor R14. An example of a suitable MOSFET M16 is the DMN601WK series MOSFET manufactured by Diodes, Inc. An example of a suitable MOSFET M17 is the NTR5103N series MOSFET manufactured by ON Semiconductor.

(54) Referring to FIG. 10, the wireless communication module **104** may be any conventional wireless interface that would enable external computing devices to communicate wirelessly with the tool **10**. Accordingly, the wireless communication module **104** may act as a wireless transceiver module for the tool **10** enabling tool **10** to transmit and receive a signal, command and/or data to and from external computing devices. In the embodiment of FIG. 10, the wireless communication module **104** uses Bluetooth® technology and/or Bluetooth® Smart technology. Preferably, the wireless communication module **104** is a Bluetooth® Low Energy (BLE) module used for communicating with a Smartphone. An example of a suitable BLE module is the HM-10 BLE module. The wireless communication module **104** is interfaced with the microcontroller **100** via a UART interface **100d** (seen in FIG. 4) embedded in the microcontroller **100**.

(55) Using Bluetooth® technology and/or Bluetooth® Smart technology, wireless communication to external computing devices is based on a serial communication protocol. As noted above, to conserve battery **20** power, the tool **10** is preferably kept in the sleep mode, which does not turn off the wireless communication module **104**. The wireless communication module **104** may continuously advertise the tool's Universally Unique Identifier (UUID). For example, the wireless communication module **104** may advertise its UUID once every 3 or 7 seconds. By continuously advertising the UUID, the wireless communication module **104** allows smartphones and other computing devices **200** to pair (or connect) with the tool **10**. Once the tool is connected with any external computing device, e.g., a smartphone, the wireless communication module **104** turns the BLUETOOTH_LED line to high, which acts as an interrupt to the microcontroller **100**. In response to the interrupt, the microcontroller **100** wakes from the sleep mode and switches the tool **10** to the active mode. The external computing devices **200** are then connected to the tool **10**, and can send commands to the microcontroller **100** to control the modules, such as the motor drive module **132** to remotely control the working head **14** of the tool **10** and/or the camera module **140** to provide a video of the working area. The wireless communication module **104** can be disabled by microcontroller **100** to conserve power using the BLUETOOTH_CTRL line, which is connected to

the sensor power control module **138**, and the MOSFET circuit comprising MOSFET **M20** and resistors **R19** and **R26**. When the **BLUETOOTH_CTRL** signal is high, the **GND**, **GND1**, and **GND2** pins are pulled to ground, thereby enabling the device. When the **BLUETOOTH_CTRL** signal is low, the **GND**, **GND1**, and **GND2** pins are not pulled to ground, thereby disabling the device. An example of a suitable MOSFET is the **DMN601WK** series MOSFET manufactured by Diodes, Inc.

(56) Configuring the BLE module may be performed using conventional AT Commands, which typically configured when the device is not paired with any external computing device. The baud rate, BLE device name, sleep mode, advertise timing, etc. can be configured using AT Commands.

(57) The USB Module **106** is described with reference to FIGS. **11A**, **11B**, and **12**.

(58) Referring to FIGS. **11A** and **11B**, the USB module **106** provides connectivity between the tool **10** and the external computing device **200**, such as a personal computer or a laptop computer (PC), as seen in FIG. **4**. The external computing device **200** may operate a User Interface Application developed in, for example, the .Net framework to communicate with the tool **100**. When connecting the tool **10** to, for example, a PC, via the USB module **106**, the tool **10** acts as virtual COM port to the PC, and communication between the tool **10** and the PC is based on conventional serial communication protocols. A micro USB connector **CON1** on the PCB of the controller **24** provides the connectivity to the PC. The USB module may also include ESD protection of 8 kilovolts, for example, to protect from the controller's PCB from external ESD surges.

(59) Referring to FIG. **12**, the controller's PCB is powered from the USB module **106**. A voltage regulator **U5**, e.g., a **TLV70033** series voltage regulator, manufactured by Texas Instruments, may be used to power the controller **24** from the USB connection point. Thus, the controller **24** may operate with USB power instead of battery **20** power. For example, data can be exchanged between the tool **10** and external computing devices, and the microcontroller **100** can monitor the operation of the tool **10** without using battery power.

(60) Referring to FIG. **13**, the motor drive module **132** is used to control the operation of the motor **18**. The motor drive module **132** uses both high side and low side switches to control the operation of the motor **18** and to limit current to the motor. The current sensor **120** was described above with reference to FIG. **7**.

(61) The microcontroller **100** controls the high side switch via MOSFET network comprising MOSFET **M22** and resistors **R20** and **R38**. Two motor control signals are provided by the microcontroller **100** via the **MOTOR_CTRL1** line which turns the MOSFET **M22** ON or OFF, as described below, to control the high side switch and the **MOTOR_CTRL2** line which turns the MOSFET **M24** ON or OFF, as described below, to control the low side switch.

(62) An example of a suitable high side switch **U10** is the **AUIR3313** Current Sense High Side Switch, manufactured by International Rectifier. The **AUIR3313** is a 4 terminal high side switch. The input signal is typically referenced to **VCC**. Thus, when the input voltage **VCC-VIN** is higher than a specified threshold V_{il} , e.g., 4.7 Volts, the output power MOSFET **M22** of the high side switch is turned ON. When the input voltage **VCC-VIN** is lower than the specified V_{il} threshold, the output power MOSFET **M22** is turned OFF. The high side switch controls the motor **18** as noted above, and is also capable of limiting the motor current. To limit the motor current, a current proportional to the power MOSFET **M22** current is sourced to the **I_{fb}** pin. An over current shutdown can occur when $V_{ifb-Vin} > 4.7V$. Either over current or over temperature latches OFF the high side switch. The current shutdown threshold can be adjusted by selecting the desired **R_{ifb}**, i.e., the current limit can be programmed by resistors **R41** & **R42**. In the exemplary embodiment of FIG. **13**, the values of **R41** (249 ohms) and **R42** (249 ohms) limits the motor current to about 87 amps. Limiting the current protects the motor **18** in the event the motor is overloaded. The high side switch **U10** can then be reset by pulling the input (**IN**) pin high.

(63) The high side switch may allow a leakage current of few micro-amps to flow through the motor **18**. To minimize the effect of this leakage current, the low side switch, MOSFET **M24** is

turned off via the MOTOR_CTRL2 line. An example of a suitable MOSFET is the CSD17559Q5 series MOSFET manufactured by Texas Instruments.

(64) Diode D7 is provided to protect the MOSFETS from the back electromagnetic force (emf) generated in the motor **18**.

(65) As described above, the tool **10** can be provided with one or more indicator modules **134** and **136**. Indicator module **134** is an LED module, and indicator module **136** is a buzzer module.

(66) Referring to FIG. **14**, an LED module **134** may be used to provide light to the area relative to the working head **14**, i.e., the working area. Preferably, ultra-bright LEDs are used for lighting the work area. Using such LEDs permits operator visibility of the working head **14** in low visibility environments. The LEDs are connected to the battery **20** power, e.g., 18V DC, and the current to the LEDs is limited using serial resistor R12. The LEDs are interfaced with the microcontroller **100** via the LED_CTRL line and a MOSFET network comprising MOSFET M28 and resistors R25 and R54. When the LED_CTRL signal is high and the trigger switch is not closed, current flows through the LEDs causing light to be emitted. When the LED_CTRL signal is low, current does not flow through the LEDs.

(67) The LED module **134** may also be used for the tool finding operation and/or to provide a visual alarm, as described below. For the tool finder operation, the microcontroller **100** can flash the LEDs by alternately switching the LED_CTRL signal high and low. Flashing the LEDs can assist in finding the tool **10**, especially in low visibility environments.

(68) Referring to FIG. **15**, a buzzer module **136** may be provided with the tool **10** of the present disclosure. In one embodiment, the buzzer module **136** may include a 95 db buzzer BZ1 that is interfaced with the microcontroller **100** via the BUZZER_CTRL line and a MOSFET network comprising MOSFET M26 and resistors R10 and R15. When the BUZZER_CTRL signal is high, current flows through the buzzer BZ1 causing the buzzer to vibrate and/or make an audible sound. When the BUZZER_CTRL signal is low, current does not flow through the buzzer BZ1. The buzzer module **136** can be used for the tool finder operation and to provide audible and tactile alarms.

(69) The working head **14** of the tool **10** is configured to perform the action for which it is designed. For example, in the embodiment shown in FIG. **1**, the working head **14** of the tool **10** is configured for performing a crimping action. When an operator initially presses the operator control **44** or **46**, the power supply module **150** activates, i.e., turns ON. Once the power supply turns ON, the microcontroller **100** starts to perform its programmed functionality, and takes control of the power usage. If no further action occurs for a period of time, i.e., the idle time expires, the microcontroller **100** turns the tool **10** power supply OFF. While the power supply is in an ON condition, the next time an operator control **44** or **46** is pressed, an interrupt provided to the microcontroller **100** causes the motor drive module **132** to actuate the motor **18** to perform the crimping action, and then stop the motor **18** after the crimping action completes.

(70) The current sensor **120**, pressure sensor **122**, battery monitor **124**, hydraulic temperature sensor **126** and PCB temperature sensor **128** are controlled by microcontroller **100** and are activated as needed. The sensed values from the sensors are monitored by the microcontroller **100** and used to control the tool **10** and/or the working head **14**. For example, if the battery voltage is below a threshold, e.g., 12.5 volts for an 18V battery, the microcontroller **100** will not enable the motor drive module **132** such that the motor **18** cannot turn ON. As another example, if the hydraulic temperature sensor **126** measures a high temperature, the microcontroller will turn OFF all the circuits and will go to Power Down Mode. Upon a successful action of the working head **14**, e.g., a successful crimp action is detected by a crimp detection algorithm, the sensed values from the sensors throughout the crimp action are stored in the flash memory **108**. These values can then be provided to the operator to analyze the performance of the tool **10** using, for example, a history function using a Smartphone and/or Windows application, as seen in FIGS. **16-18**. The RTCC **110** provides a date and timestamp to each record stored in the flash memory **108**.

(71) The microcontroller **100** is connected to motor drive module **132** (seen in FIG. **13**) and controls the operation of the motor **18**. For example, the microcontroller **100** can turn the motor **18** ON and OFF, and can control the current to the motor **18** protect the motor from overload conditions. The microcontroller **100** is connected to one or more indicator modules. LED module **134** and buzzer module **136** can be used to provide an alarm or otherwise notify operators that an abnormal condition exists with the tool **10**, such as misuse, high temperature, low battery power, etc. For example, the microcontroller **100** can be connected to a buzzer module **136** that provides an audible or tactile alarm that an abnormal condition exists with the tool **10**. As another example, the microcontroller **100** can be connected to an LED module **134** that provides a visible alarm (e.g., a sequence of flashing lights) that an abnormal condition exists. The LED module may also be configured by the microcontroller to illuminate a working area, i.e., an area where the working head **14** of the tool is operating. Illuminating the working area may enhance the video of the working area taken by the camera **26**.

(72) Referring to FIGS. **16-20**, the wireless communication module **104** is connected to the microcontroller **100** via a UART interface **100d** embedded in the microcontroller. The wireless communication module **104** permits wireless communication between the tool **10** and external computing devices **200** with a wireless transceiver included in such external computing devices. The wireless transceiver included in such external computing devices **200** may also be referred to herein as the device wireless transceiver. Non-limiting example of the external computing devices include smartphones, mobile phones, tablets, personal digital assistant (PDA), laptop computer, navigation device, portable computing unit, game console, desktop computer system, workstation, Internet appliance and the like, using for example conventional communication protocols. For example, the wireless communication module **104** may be configured for Bluetooth® communication with a Bluetooth® enabled device wireless transceiver typically included at least in the computing devices noted above. Computing devices using wireless communications provide the microcontroller **100** an interrupt whenever the computing device connects to the tool **10**. This enables the microcontroller **100** to wake up from sleep mode and respond to the connected computing device. Computing devices with USB connectivity can be connected to the tool **10** via USB module **106** which is connected to a UART interface **100d** embedded in the microcontroller **100**. Preferably, the wireless communication interface **104** provides communication to portable computing devices such as smartphones and tablets, while the USB module **106** provides communication less portable computing devices, such to personal computers or laptop computers. The tool **10** can provide all information collected from the Sensors, Modules and derived from the microcontroller to operators through the wireless communication module **104** and/or USB module **106**. Examples of such information include total working head action count, e.g., total crimp count, total motor **18** runtime, available battery power, etc.

(73) Referring to FIGS. **16-18** for example, the tool **10** can be connected to a smartphone **200** that can then be used to obtain information from the tool or remotely control the operation of the tool. Referring to the exemplary embodiment of FIG. **16**, the smartphone can include an application that when launched presents a display **204** that shows a state of the connection between the smartphone **200** and the tool **10**, the serial number of the tool and the manufacturing date of the tool. The smartphone **200** can also display a main menu of tool command operations that can be used to send commands to the controller **24** to retrieve tool related information or control the operation of the motor **18**. For example, selecting the FIND TOOL command sends a command to the controller **24** to activate a tool finder operation described below. Selecting the TOOL STATUS command causes the display **204** to present tool status information as shown in FIG. **17**. In this exemplary embodiment, the tool status information shown includes current date, current time, number of working head actions performed by the tool, e.g., crimp actions, motor run time, and battery status. Selecting the back command returns to the main menu. Selecting the HISTORY command causes the display **204** to present a history of the working head actions performed by the tool, e.g., crimp

actions, as shown in FIG. 18. The information presented on the display 204 of the history of the working head actions may also be referred to as working head operation information. Selecting the back command returns to the main menu. Selecting the MOTOR RUN command causes the smartphone to send a command to the controller 24 that causes the microcontroller 100 to activate the motor drive module 132 as described above.

(74) Selecting the VIDEO command causes the display 204 to present a video window and one or more video and tool related operation buttons or icons, as shown in FIG. 19. For example, selecting the PLAY button would send a command to the controller 24 causing the microcontroller 100 to activate the camera 26 and to feed the video image information from the camera to the smartphone via the wireless communication module 104, as shown in FIG. 20. While watching the video, an operator can then select the MOTOR RUN button. Selecting the MOTOR RUN button would operate as described above. Selecting the LIGHT button would send a command to the controller 24 causing the microcontroller 100 to activate the LED module. Selecting the back command returns to the main menu.

(75) Using the wireless connectivity capability of the tool 10 allows for Tool Finder operations and Remote Operation of the tool.

(76) With the tool finder operation, a smartphone or other computing device 200 connected to the tool 10 can send a signal to the microcontroller 100 to initiate the tool finder operation. The tool finder operation could be to activate the LED module 134 to provide a visible indication where the tool is located, or the tool finder operation could activate the buzzer module 136 to provide an audible indication where the tool is located, or the tool finder operation can be to activate both the LED module 134 and the buzzer module 136.

(77) For remote operation of the tool 10, a smartphone or other external computing device connected to the tool 10 can send a signal, i.e., commands, to the microcontroller 100 to initiate remote operation of the tool 10. Commands from a smartphone application can cause the microcontroller 100 to initiate the action of the working head 14 of the tool 10. For example, if the tool 10 is a crimping tool, a smartphone can send a command to the microcontroller 100 to initiate the crimp action of the tool 10 remotely. Remote operation provides operators with the ability to perform the working head action, e.g., crimp action in hazardous situations, etc. The tool 10 can also provide security against unauthorized access to the remote operation with an operator defined password.

(78) As described above, the tool according to the present disclosure can include any type of working head. Examples of such working heads can be found on the PATMD-LLI Series of in-line pole cutters, manufactured by Burndy®, the PAT46LWS-LI Scoop Style C-Head Crimper, manufactured by Burndy®, the PAT46LW-LI Latch Head Crimper, manufactured by Burndy®, the PAT750-LI C-Head Crimper, manufactured by Burndy®, the PATCUT2156-LI, PATCUT245-LI and PATCUT129-LI Latch Head Cutters, manufactured by Burndy®. It will be understood that various modifications can be made to the embodiments of the present disclosure herein without departing from the spirit and scope thereof. Therefore, the above description should not be construed as limiting the disclosure, but merely as embodiments thereof. Those skilled in the art will envision other modifications within the scope and spirit of the invention as defined by the claims appended hereto.

Claims

1. A portable tool comprising: a frame; a working head operatively coupled to the frame and configured to perform a crimping or cutting operation; a motor positioned within the frame, a hydraulic system positioned within the frame and operatively coupled to the motor, and a ram operatively coupled to the hydraulic system and movably connected to the working head; a camera attached to the frame so that the camera is focused on a working area of the working head; a

controller positioned within the frame and including a motor drive module configured to control an operation of the motor, and a wireless transceiver module interactive with the controller and configured to transmit tool information to an external computing device, and to receive one or more commands from the external computing device to activate the motor drive module causing the motor to activate the hydraulic system to cause the ram to initiate the crimping or cutting operation or to deactivate the motor drive module.

2. The portable tool according to claim 1, wherein the frame includes an indicator that is selectively controlled by the controller such that when the controller receives a tool finder command from the wireless transceiver module the controller activates the indicator.

3. The portable tool according to claim 2, wherein the indicator comprises an audible indicator, a visual indicator, or audible and visual indicators.

4. The portable tool according to claim 1, wherein the camera is interactive with the controller, and wherein the tool information includes one or more images of the working area of the working head generated by the camera that can be transmitted to the external computing device.

5. The portable tool according to claim 4, wherein the one or more images include video of the working area of the working head.

6. The portable tool according to claim 1, wherein the tool information comprises at least one of tool status information and working head operation history information.

7. A portable tool system comprising: an external computing device having a device wireless transceiver configured to receive transmitted tool information and send one or more commands; and a portable tool including: a frame; a working head operatively coupled to the frame and configured to perform a crimping or cutting operation; a motor positioned within the frame, a hydraulic system positioned within the frame and operatively coupled to the motor, and a ram operatively coupled to the hydraulic system and movably connected to the working head; and a controller positioned within the frame and including a motor drive module configured to control an operation of the motor, and a wireless transceiver module capable of transmitting the tool information or receiving the one or more commands; wherein the one or more commands transmitted by the external computing device are received by the wireless transceiver module causing the controller to activate or deactivate an operation of the portable tool; and wherein the tool information transmitted by the portable tool is received by the external computing device such that the tool information can be presented to an operator of the external computing device.

8. The portable tool system according to claim 7, wherein the one or more commands cause the controller to activate or deactivate the motor drive module.

9. The portable tool system according to claim 7, wherein the frame includes an indicator that is selectively controlled by the controller such that when the controller receives a tool finder command from the wireless transceiver module the controller activates the indicator.

10. The portable tool system according to claim 9, wherein the indicator comprises an audible indicator, a visual indicator, or audible and visual indicators.

11. The portable tool system according to claim 7, wherein the portable tool includes a camera attached to the frame and interactive with the controller, and wherein the tool information includes one or more images of the working area of the working head generated by the camera that can be transmitted to the external computing device.

12. The portable tool system according to claim 11, wherein the one or more images include video of the working area of the working head.

13. The portable tool system according to claim 7, wherein the tool information comprises at least one of tool status information and working head operation history information.

14. A portable tool comprising: a frame; a working head operatively coupled to the frame and configured to perform a crimping or cutting operation; a motor positioned within the frame, a hydraulic system positioned within the frame and operatively coupled to the motor, and a ram operatively coupled to the hydraulic system and movably connected to the working head; and a

controller positioned within the frame and including a motor drive module configured to control an operation of the motor, and a wireless transceiver module capable of transmitting tool information or receiving one or more commands; wherein the one or more commands transmitted by an external computing device is received by the wireless transceiver module causing the controller to activate or deactivate an operation of the portable tool; and wherein the tool information transmitted by the portable tool can be received by the external computing device such that the tool information can be presented to an operator of the external computing device.

15. The portable tool system according to claim 14, wherein the one or more commands cause the controller to activate or deactivate the motor drive module.

16. The portable tool system according to claim 14, wherein the frame includes an indicator that is selectively controlled by the controller such that when the controller receives a tool finder command from the wireless transceiver module the controller activates the indicator.

17. The portable tool system according to claim 16, wherein the indicator comprises an audible indicator, a visual indicator, or audible and visual indicators.

18. The portable tool system according to claim 14, wherein the tool includes a camera that is interactive with the controller, and wherein the tool information includes one or more images of the working area of the working head generated by the camera that can be transmitted to the external computing device.

19. The portable tool system according to claim 18, wherein the one or more images include video of the working area of the working head.

20. The portable tool system according to claim 14, wherein the tool information comprises at least one of tool status information and working head operation history information.
